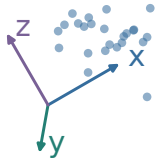
Registration and three-layer scene preparation

a Positive Sim3 and an explicit world frame



source frame

$$x_w = sRx_s + t$$

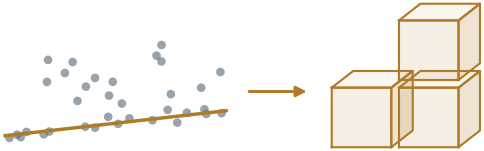
Z-up world

$s > 0, \quad R^\top R = I$

Scale needs evidence

else UNVERIFIED

b Bounded collision estimates



floor: $n^\top x + d = 0$

sample ≤100k · proxies ≤2k

defaults; coarsen to meet cap

c Three runtime layers

Appearance

source Gaussians

Collision

static proxies

Interaction

explicit objects

Source appearance, estimated collision and declared objects remain distinct.